A Report on

DEEPINSPECT AI

AI DRIVEN NETWORK SECURITY SYSTEM

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Declaration

We hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Certificate

Certified that **Shashank Kumar Srivastava, Aryan Gupta, Ankit Raj,** has carried out the project work presented in this report entitled **"DEEPINSPECT AI"** for the award of **Bachelor of Technology** from Inderprastha Engineering College, Ghaziabad, under my supervision. The report embodies result of original work and studies carried out by Student himself and the contents of the report do not form the basis for the award of any other degree to the candidate or to anybody else**.**

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Abstract

In the rapidly evolving landscape of cybersecurity, traditional rule-based intrusion detection and prevention systems are increasingly inadequate in identifying and responding to the sophisticated and dynamic threats posed by modern cyber attackers. Static systems, while useful for known threat signatures, often struggle with detecting novel or subtle malicious behaviours that evolve over time. This research introduces the design, implementation, and evaluation of an AI-Driven Network Security System (AIDNSS) that harnesses the capabilities of machine learning and data-driven modelling to proactively secure digital network environments. The proposed system combines real-time traffic monitoring with advanced anomaly detection techniques to uncover suspicious behaviour that may indicate the presence of cyber threats such as Distributed Denial of Service (DDoS) attacks, port scanning activities, malware communications, and data exfiltration attempts. Leveraging both supervised and unsupervised learning algorithms, including decision trees, support vector machines, and clustering models like K-means and DBSCAN, the system continuously learns from network data to improve its detection accuracy over time. This adaptive approach reduces the reliance on manually defined rules and signature databases, enabling the system to evolve alongside emerging

threat patterns without constant human oversight. Furthermore, the system incorporates intelligent automation to not only detect potential intrusions but also initiate timely response actions—such as alert generation, traffic blocking, or policy updates—thereby minimizing the window of vulnerability. The research includes the deployment and testing of the system in a controlled, simulated network environment to evaluate its performance against standard attack scenarios. Metrics such as detection rate, false positive rate, response time, and resource utilization were analysed to benchmark the system against conventional security solutions. Experimental results show a significant improvement in threat detection capabilities, including a marked reduction in false positives and enhanced responsiveness, affirming the potential of AI-powered approaches in next-generation network security architectures. Ultimately, this work seeks to bridge the gap between the static nature of legacy security mechanisms and the fluid, increasingly intelligent landscape of cyber threats, emphasizing the role of machine learning as a cornerstone for future cybersecurity solutions.

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